



Trigonometric Function

Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

1 Find the exact value of $\cos \frac{5\pi}{6}$. 1

$$\cos \frac{5\pi}{6} = -\cos \frac{\pi}{6} = -\cos 30^\circ = -\frac{\sqrt{3}}{2}$$

2 Show that $\frac{(1+\tan^2 \theta)\cot \theta}{\operatorname{cosec}^2 \theta} = \tan \theta$. 2

$$\begin{aligned} \text{LHS} &= \frac{(1+\tan^2 \theta)\cot \theta}{\operatorname{cosec}^2 \theta} \\ &= \frac{\sec^2 \theta \cot \theta}{\operatorname{cosec}^2 \theta} \\ &= \frac{1}{\cos^2 \theta} \times \frac{\cos \theta}{\sin \theta} \div \frac{1}{\sin^2 \theta} \\ &= \frac{1}{\cos \theta \sin \theta} \times \sin^2 \theta = \frac{\sin \theta}{\cos \theta} = \tan \theta = \text{RHS} \end{aligned}$$

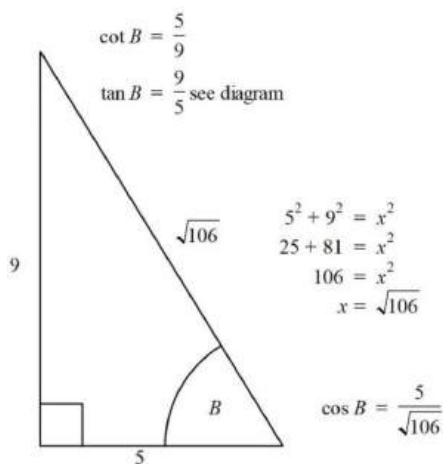
3 What is the solution to the equation $2\cos x = \sqrt{3}$ for x , where $0 \leq x \leq 2\pi$? 1

$$\begin{aligned} 2\cos x &= \sqrt{3} \\ \cos x &= \frac{\sqrt{3}}{2} \\ x &= \frac{\pi}{6} \text{ and } \frac{11\pi}{6} \end{aligned}$$

- 4 The minute hand on a clock face is 12 centimetres long. 1
- a. What angle does the hand move in 40 minutes? Answer in radians. 1
- b. How far does the tip of the hand move in 40 minutes? 1
- c. What area does the hand sweep through in 40 minutes? 1

$$\begin{aligned} \text{(a)} \quad \text{Angle} &= \frac{40}{60} \times 360^\circ = \frac{40}{60} \times 2\pi \\ &= \frac{4\pi}{3} \text{ radians} \\ \text{(b)} \quad l &= r\theta = 12 \times \frac{4\pi}{3} = 16\pi \text{ cm} \\ \text{(c)} \quad A &= \frac{1}{2} r^2 \theta = \frac{1}{2} \times 12^2 \times \frac{4\pi}{3} = 96\pi \text{ cm} \end{aligned}$$

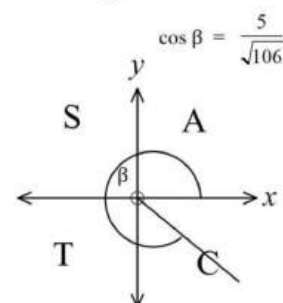
5 Given that $\cot \beta = -\frac{5}{9}$ and that $\sin \beta < 0$, find the value of $\cos \beta$. 2



$\cot$ (and $\tan$) are -ve
 so 2nd or 4th Quad

$\sin$ is -ve
 so 3rd or 4th Quad

So 4th Quad so $\cos$ +ve



6 Find the exact circular measure (in radians) for the following angles.

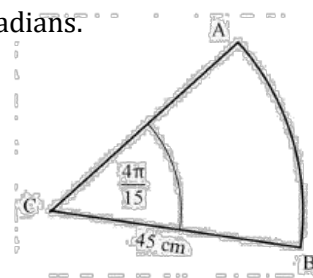
a.	75°	a)	$75^\circ = \frac{75\pi}{180} = \frac{5\pi}{12}$	1
b.	228°	b)	$228^\circ = \frac{228\pi}{180} = \frac{19\pi}{15}$	1

7 A sector of a circle is shown below, with the internal angle measured in radians.

a. Calculate the exact length of the arc AB .

b. Find the exact area of the sector ABC .

a)	$l = r\theta$ $= 45 \times \frac{4\pi}{15} = 12\pi \text{ cm}$	1
b)	$\text{Area} = \frac{1}{2}r^2\theta$ $= \frac{1}{2} \times 45^2 \times \frac{4\pi}{15} = 270\pi \text{ cm}^2$	1

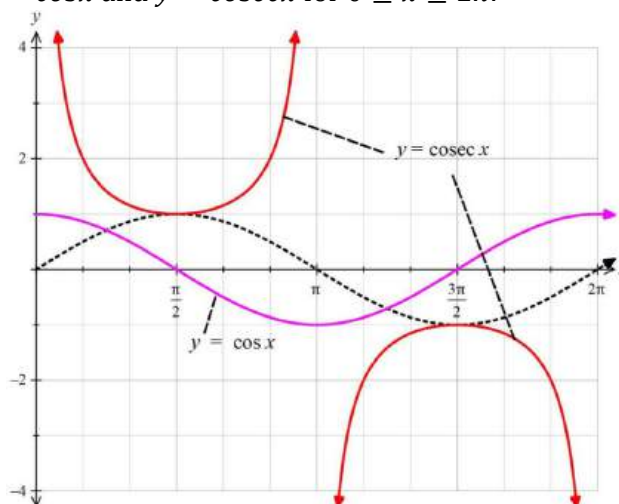


8 a. Express $\sin(180 - \alpha) - \sin(360 - \alpha)$ in terms of $\sin\alpha$.

b. Show that $2\sec^2\theta + \frac{\sin^2\theta - 2\cos^2\theta}{\cos^2\theta} = 3\tan^2\theta$.

a)	$\sin(180 - \alpha) - \sin(360 - \alpha) = \sin\alpha - (-\sin\alpha)$ $= \sin\alpha + \sin\alpha = 2\sin\alpha$	1
b)	$2\sec^2\theta + \frac{\sin^2\theta - 2\cos^2\theta}{\cos^2\theta} = 2\sec^2\theta + \frac{\sin^2\theta}{\cos^2\theta} - \frac{2\cos^2\theta}{\cos^2\theta}$ $= 2\sec^2\theta + \tan^2\theta - 2$ $= 2\sec^2\theta - 2 + \tan^2\theta$ $= 2\tan^2\theta + \tan^2\theta = 3\tan^2\theta$	2

9 The graph of $y = \sin x$ for $0 \leq x \leq 2\pi$ is drawn on the axes below. On the same axes, draw and label neat sketches of $y = \cos x$ and $y = \csc x$ for $0 \leq x \leq 2\pi$.



10 What is solution to the equation $2\cos\beta = -\sqrt{3}$ for $0^\circ \leq \beta \leq 360^\circ$?

$$\cos\beta = -\frac{\sqrt{3}}{2}$$

Exact value is 30 in first quadrant, but we have a negative result.

Cos is negative in quadrant 2 & 3

Therefore, $(180-30)$ and $(180+30) = 150$ or 210

- 11 What is the solution to the equation $\cos\left(\frac{\theta}{2} + 30^\circ\right) = \sin\theta$ for $0^\circ \leq \theta \leq 90^\circ$? 1

Sine and cosine are complementary angles: $\sin\theta = \cos(90 - \theta)$

$$\cos\left(\frac{\theta}{2} + 30^\circ\right) = \sin\theta$$

$$\frac{\theta}{2} + 30 = 90 - \theta$$

$$\theta + 60 = 180 - 2\theta \text{ or } 3\theta = 120 \text{ or } \theta = 40^\circ$$

- 12 In the diagram, AB is an arc of a circle with centre O .

The radius $OA = 12 \text{ cm}$ and $\angle AOB = \frac{5\pi}{6}$ radians.

- a. Find the exact value of the length of the arc AB .

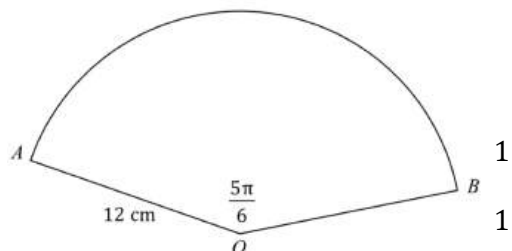
- b. Find the area of sector AOB .

$$l = r\theta$$

$$= 12 \times \frac{5\pi}{6} = 10\pi \text{ cm}$$

$$l = \frac{1}{2}r^2\theta$$

$$= \frac{1}{2} \times 12^2 \times \frac{5\pi}{6} = 60\pi \text{ cm}^2$$



- 13 Prove that $1 + \tan^2 \theta = \frac{1}{1 - \sin^2 \theta}$. 2

$$\text{LHS} = 1 + \tan^2 \theta = \sec^2 \theta = \frac{1}{\cos^2 \theta} = \frac{1}{1 - \sin^2 \theta} = \text{RHS}$$

- 14 It is given that $\cos\alpha = \frac{1}{3}$ for $270^\circ < \alpha < 360^\circ$. What is the value of $\tan\alpha$? 1

$$\sin\alpha = -\sqrt{1 - \cos^2\alpha} = -\frac{\sqrt{8}}{3}$$

$$\tan\alpha = \frac{\sin\alpha}{\cos\alpha} = -\sqrt{8} = -2\sqrt{2}$$

- 15 How many solutions does the equation $\sin 2x = 0$ have for $0 \leq x \leq \pi$? 1

$$2x = \pi k \Rightarrow x = \frac{\pi k}{2} \text{ for } k = 0, 1, 2$$

- 16 a. Solve $2\sin^2 x - \sin x = 0$ for $0 \leq x \leq 2\pi$. 2

$$\sin x(2\sin x - 1) = 0$$

$$\sin x = 0 \Rightarrow x = 0, \pi, 2\pi$$

$$\sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}, \frac{5\pi}{6}$$

- b. Show that $\frac{\sin x + 1}{\cos x + \cot x} = \tan x$.

$$\frac{\sin x + 1}{\cos x + \cot x}$$

$$= \frac{\sin x + 1}{\cos x + \frac{\cos x}{\sin x}}$$

$$= \frac{\sin x + 1}{\frac{\cos x \sin x + \cos x}{\sin x}}$$

$$= \frac{\sin^2 x + \sin x}{\cos x \sin x + \cos x}$$

$$= \frac{\sin x (\sin x + 1)}{\cos x (\sin x + 1)}$$

$$= \frac{\sin x}{\cos x}$$

$$= \tan x$$

Trigonometric Function

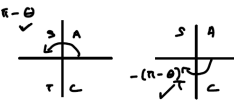
Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

- 1 Solve $2\cos\theta + 1 = 0$ for the domain $[-\pi, \pi]$.

3

$$\begin{aligned} 2\cos\theta + 1 &= 0, & -\pi \leq \theta \leq \pi \\ 2\cos\theta &= -1 \\ \cos\theta &= -\frac{1}{2} \end{aligned}$$


$$\theta = \cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$$

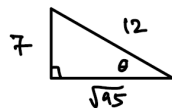
$$\therefore \theta = \pi - \frac{\pi}{3}, -\left(\pi - \frac{\pi}{3}\right)$$

$$\theta = \frac{2\pi}{3}, -\frac{2\pi}{3}$$

- 2 Given $\sin\theta = \frac{7}{12}$ for an acute angle θ , find the exact value of $\tan\theta$.

2

$$\sin\theta = \frac{7}{12}$$



$$\therefore \tan\theta = \frac{O}{A} = \frac{7}{\sqrt{5}}$$

- 3 Prove that $\frac{\tan A}{1+\sec A} - \frac{\tan A}{1-\sec A} = 2\operatorname{cosec} A$.

$$\text{LHS} = \frac{\tan A}{1+\sec A} - \frac{\tan A}{1-\sec A}$$

3

$$= \frac{\tan A(1-\sec A) - \tan A(1+\sec A)}{(1+\sec A)(1-\sec A)}$$

$$= \frac{\cancel{\tan A} - \tan A \sec A - \cancel{\tan A} - \tan A \sec A}{1 - \sec^2 A}$$

$$= \frac{-2 \tan A \sec A}{-\tan^2 A}$$

$$= \frac{2 \times \frac{1}{\cos A}}{\frac{\sin}{\cos A}} = 2 \operatorname{cosec} A = \text{RHS}$$

- 4 A vehicle has wheels with a diameter of 80 cm (including the tyre) that are spinning at a rate of 2.6 radians per second. How far will the vehicle travel in one minute?

2

$$\frac{d\theta}{dt} = 2.6 \text{ rad/s} \quad \begin{aligned} 1 \text{ s} &= 2.6 \text{ rad} \\ 60 \text{ s} &= 156 \text{ rad} \end{aligned}$$

$$l = r \times \theta$$

$$2\pi \text{ rad} = 2\pi r \quad (1 \text{ circumference})$$

$$2\pi \text{ rad} = 2\pi \times 40 = 80\pi \text{ cm}$$

$$1 \text{ rad} = 40 \text{ cm}$$

$$156 \text{ rad} = 6240 \text{ cm} = \boxed{62.4 \text{ m}}$$



- 5 Consider the diagram shown. What is the area of the shaded segment?

2

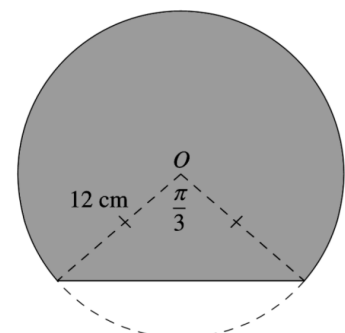
$$A = \pi r^2 - \frac{1}{2} r^2 (\theta - \sin\theta)$$

$$= \pi \times 12^2 - \frac{1}{2} \times 12^2 \left(\frac{\pi}{3} - \sin \frac{\pi}{3} \right)$$

$$= 144\pi - 72 \left(\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right)$$

$$= 144\pi - 24\pi + 36\sqrt{3}$$

$$= 120\pi + 36\sqrt{3}$$



6 Solve $\sin^2 2x = \frac{1}{4}$ for $-180^\circ \leq x \leq 180^\circ$.

3

$$\sin^2 2x = \frac{1}{4}$$

$$\sin 2x = \pm \frac{1}{2} \quad , \quad -360 \leq 2x \leq 360^\circ$$

$$2x = \sin^{-1}\left(\frac{1}{2}\right) = 30^\circ$$

$$\therefore 2x = 30^\circ, 150^\circ, 210^\circ, 330^\circ, \\ -30^\circ, -150^\circ, -210^\circ, -330^\circ$$

$$\therefore x = 15^\circ, 75^\circ, 105^\circ, 165^\circ, \\ -15^\circ, -75^\circ, -105^\circ, -165^\circ$$

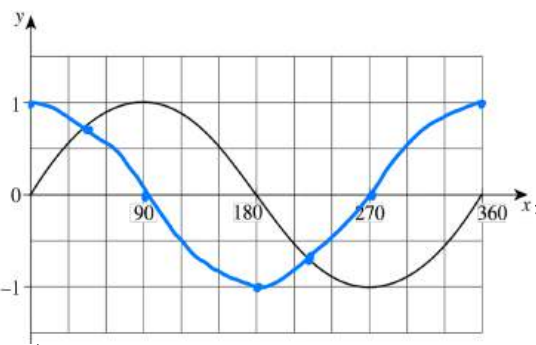
7 What is the range of $3\cos 2x$?

Range: $-3 \leq y \leq 3$

1

$$\therefore [-3, 3]$$

8 The number plane shows the graph of $y = \sin x^\circ$ for $[0, 360]$.



a. On the number plane, draw the graph of

$y = \cos x^\circ$ for $[0, 360]$.

2

b. Hence, solve $\sin x^\circ = \cos x^\circ$ for $[0, 360^\circ]$.

From the diagram,

$$\sin x^\circ = \cos x^\circ$$

when $x = 45^\circ, 225^\circ$

1

9 What is 120° in radians?

$$120^\circ \times \frac{\pi}{180} = \frac{2\pi}{3}$$

1

10 If the angle of a sector with radius 1 unit is 18° , what is the area of the sector?

2

$$\theta = 18^\circ \times \frac{\pi}{180} = \frac{\pi}{10}$$

$$A = \frac{1}{2} r^2 \theta = \frac{1}{2} \times 1^2 \times \frac{\pi}{10}$$

$$= \frac{\pi}{20} \text{ unit}^2$$

11 Let $f(x) = \cos x$.

a. Show that $f(x)$ is an even function.

$$(\because) f(-x) = \cos(-x) = \cos x$$

$$f(-x) = f(x)$$

$\therefore f(x)$ is an even function.

1

b. Write down the domain and range of $f(x)$. $D: (-\infty, \infty)$

$R: [-1, 1]$

2

c. Explain why the function is **not** one-to-one.

2

The function is not one-to-one because it doesn't pass the horizontal line test.
i.e. Some horizontal line intersects the graph of the function more than once.

e.g. when $y = \frac{1}{2}$: $\frac{1}{2} = \cos x$
 $x = -\frac{\pi}{3}, \frac{\pi}{3}, \frac{5\pi}{3}, \dots$

12

a. Show that $(1 + \tan x) \times \frac{\sin x}{\sin x + \cos x} = \tan x$.

$$\begin{aligned}
 \text{LHS} &= (1 + \tan x) \cdot \frac{\sin x}{\sin x + \cos x} \\
 &= \left(1 + \frac{\sin x}{\cos x}\right) \times \frac{\sin x}{\sin x + \cos x} \\
 &= \frac{\cos x + \cancel{\sin x}}{\cos x} \times \frac{\sin x}{\cancel{\sin x} + \cos x} \\
 &= \frac{\sin x}{\cos x} = \tan x = \text{RHS} \quad \neq
 \end{aligned}$$

2

b. Hence, or otherwise, simplify $\frac{(\sin 2x + \cos 2x) \operatorname{cosec} 2x}{1 + \tan 2x}$.

1

$$\frac{(\sin 2x + \cos 2x) \operatorname{cosec} 2x}{1 + \tan 2x} = \frac{(\sin 2x + \cos 2x)}{(1 + \tan 2x) \sin 2x} = \frac{1}{\tan 2x} = \boxed{\cot 2x}$$